

Chapter 3

Demand, Supply, and Market Equilibrium

Lecturer: Mihyun You

Department of Economics
Georgetown University

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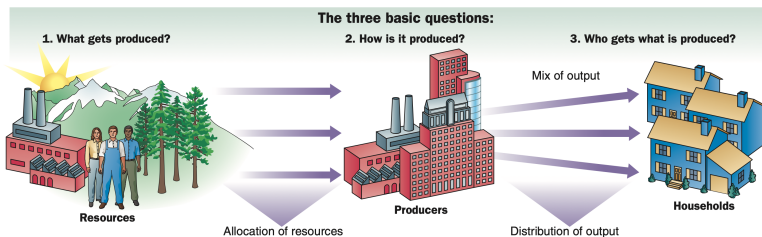
- ① Basic Economic Agents
- ② Supply and Demand
- ③ Market Equilibrium



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The Economic Problem

- Economics is essentially a study of *choice* in a world of scarcity.
- The origin of economic problems is ascribed to the fact that human wants are infinite but resources are finite.
- Every economy (society) faces following questions:
 - ① What and how much to produce?
 - ② How to produce?
 - ③ For whom to produce?



▲ **FIGURE 2.1** The Three Basic Questions

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The Basic Decision-Making Units

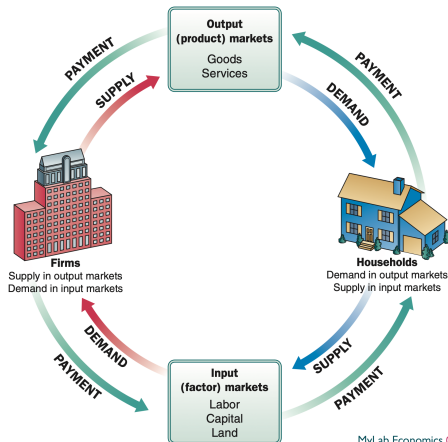
Firms and Households

- A market economy is a system in which market plays a decisive role in solving economic problems, without any central planning or direction from governmental institutions.
- Firms and households are the basic forces at work in market systems.
 - Firms: primary producing units
 - Households: primary consuming units
- Firms make decisions to maximize profits by producing goods or services with the given amount of resources.
- Households make decisions to maximize utilities by consuming goods or services with the given amount of income and wealth.

The Circular Flow

Input Markets and Output Markets

Input and output markets are connected through the behavior of both firms and households.



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Demand

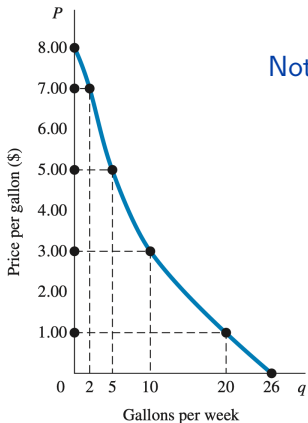
- Quantity demanded: The amount (number of units) of a product that a household would buy in a given period if it could buy all it wanted at the current market price.

TABLE 3.1 Alex's Demand Schedule for Gasoline

Price (per Gallon)	Quantity Demanded (Gallons per Week)
\$ 8.00	0
7.00	2
6.00	3
5.00	5
4.00	7
3.00	10
2.00	14
1.00	20
0.00	26

Demand Curve

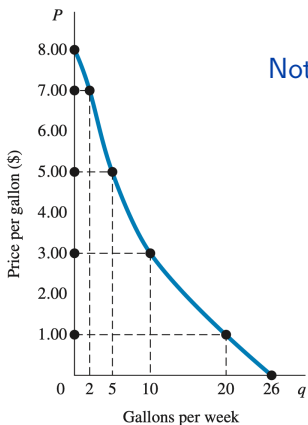
A demand curve graphically illustrates a demand schedule.



Note: Properties of Demand Curves

Demand Curve

A demand curve graphically illustrates a demand schedule.



Note: Properties of Demand Curves

- Law of demand: demand curves have a negative slope.
- X-intercept: a result of time limitations and diminishing marginal utility.
- Y-intercept: a result of limited income and wealth.

Changes in Quantity Demanded vs. Changes in Demand

- Changes in price of the product affect the *quantity demanded*.
- Changes in any other factor—such as incomes & wealth, preferences, expectations, and price of related goods—affect *demand*.

Note: Income and Demand

- Normal goods: income $\uparrow \rightarrow$ demand $\uparrow$
- Inferior goods: income $\uparrow \rightarrow$ demand $\downarrow$

Note: Related Goods and Demand

- Substitutes: price of one good $\uparrow \rightarrow$ demand for the other good $\uparrow$
- Complements: price of one good $\uparrow \rightarrow$ demand for the other good $\downarrow$

Exercise

! Exercise: Draw a graph on a (q, P) -plane representing the effect of an increase in income on gasoline demand.

TABLE 3.2 Shift of Alex's Demand Schedule Resulting from an Increase in Income

Price (per Gallon)	Schedule D_0	Schedule D_1
	Quantity Demanded (Gallons per Week at an Income of \$500 per Week)	Quantity Demanded (Gallons per Week at an Income of \$700 per Week)
\$ 8.00	0	3
7.00	2	5
6.00	3	7
5.00	5	10
4.00	7	12
3.00	10	15
2.00	14	19
1.00	20	24
0.00	26	30

Supply

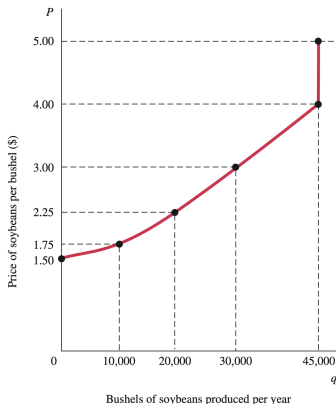
- Quantity supplied: The amount of a particular product that a firm would be willing and able to offer for sale at a particular price during a given time period.

TABLE 3.3 Clarence Brown's Supply Schedule for Soybeans

Price (per Bushel)	Quantity Supplied (Bushels per Year)
\$ 1.50	0
1.75	10,000
2.25	20,000
3.00	30,000
4.00	45,000
5.00	45,000

Supply Curve

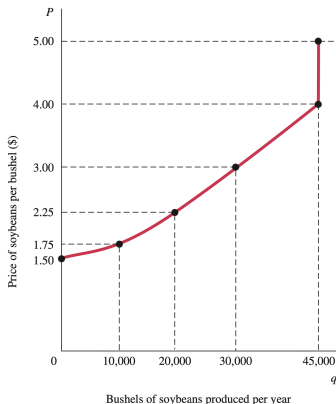
A supply curve graphically illustrates a supply schedule.



Note: Properties of Supply Curves

Supply Curve

A supply curve graphically illustrates a supply schedule.



Note: Properties of Supply Curves

- Law of supply: Short-run supply curves have a positive slope.
- In the long run, supply curve exhibit a vertical line.

Changes in Quantity Supplied vs. Changes in Supply

- Changes in price of the product affect the *quantity supplied*.
- Changes in any other factor—such as production cost, technologies, and price of related products—affect *supply*.

Note: Related Products and Supply

- Substitutes: price of one good $\uparrow \rightarrow$ supply of the other good $\downarrow$
- Byproducts: price of one good $\uparrow \rightarrow$ supply of the other good $\uparrow$

Exercise

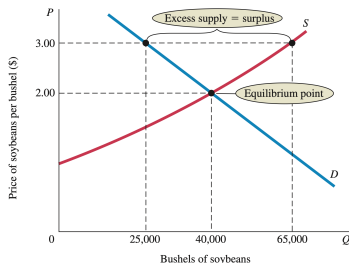
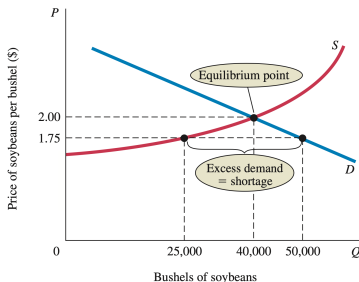
! Exercise: Draw a graph on a (q, P) -plane representing the effect of the development of a new disease-resistant seed strain on soybean supply.

TABLE 3.4 Shift of Supply Schedule for Soybeans following Development of a New Disease-Resistant Seed Strain

Price (per Bushel)	Schedule S_0	Schedule S_1
	Quantity Supplied (Bushels per Year Using Old Seed)	Quantity Supplied (Bushels per Year Using New Seed)
\$ 1.50	0	5,000
1.75	10,000	23,000
2.25	20,000	33,000
3.00	30,000	40,000
4.00	45,000	54,000
5.00	45,000	54,000

Market Equilibrium

- Now, we are ready to see how supply and demand interact to determine the final market equilibrium price and quantity.



- The price adjustment process continues until when an equilibrium price is reached, at which point the quantity supplied and quantity demanded are equal.

Market Equilibrium with Equations

- Assume that the equation of the inverse demand curve and the inverse supply curve are as follows:

$$P = a - bQ_d,$$

$$P = c + dQ_s$$

where Q_d , Q_s and P denote quantity demanded, quantity supplied, and price, respectively.

- Then the demand curve will be

$$Q_d = \frac{a}{b} - \frac{1}{b}P.$$

- The supply curve will be

$$Q_s = \frac{c}{d} + \frac{1}{d}P.$$

Computation for Market Equilibrium

- In the graphical illustration, we find the equilibrium price and quantity by finding the point of intersection of the supply and demand curves.
 - For mathematical computation, we can simply set Q_d equal to Q_s and call it Q^* , which gives us two equations in two unknowns (Q^*, P^*) .
- ! Exercise: End-of-Chapters practice problems 5.5 in the textbook